a finite habitat patch (e.g., Kot and Schaffer 1986; Van Kirk and Lewis 1997; Reimer et al. 2016), and critical speeds of climate-change-driven range shifts (e.g., Zhou and Kot 2011; Harsch et al. 2014; Cobbold and Stana 2020).

IDEs are spatially nonlocal models (Lee et al. 2001) that are discrete in time and continuous in space. They are especially useful for modeling the dynamics of species that have non-overlapping generations and distinct growth and dispersal stages (Kot and Schaffer 1986; Lutscher 2019). The basic IDE for a population with discrete and nonoverlapping generations is

$$n_{t+1}(x) = \int_{\Omega} k(x, y)\, g[n_t(y)]\, dy, \tag{1}$$

where $n_t(x)$ is the population density in generation t at a spatial location x, Ω is the spatial domain, $k(x, y)$ is the dispersal kernel, and $g[n_t(y)]$ is the growth or recruitment function. Population density is mapped from the current generation to the next in two distinct phases. First, the population grows, while remaining fixed in space. In particular, $g[n_t(y)]$ gives the new population-density at location y after growth has occurred. After growth, individuals disperse, relocating from starting location y to new location x with probability governed by $k(x, y)$. The dispersal kernel is a probability density function for the final location x of the individuals.

In contrast to reaction–diffusion models, which are continuous in both time and space, IDE models account for seasonal growth and can incorporate a variety of dispersal patterns, including long-distance dispersal (Lewis et al. 2016). The ability to choose a dispersal kernel that best fits the population being modeled is a key feature of IDEs. As a result, IDE models have the advantage of increased ecological realism, as many species possess the features of seasonal growth and long-distance dispersal (Okubo and Levin 1989; Andersen 1991; Kot et al. 1996; Lutscher 2019).

Despite the advantages of IDEs, these models have drawbacks as well. IDEs can be computationally expensive since, as for most spatially nonlocal models, individuals may disperse long distances (Lutscher 2019). At the extreme end of this spatial complexity, under certain conditions, a propagule may start from any location in the spatial domain of the model and disperse to any other location in that domain.

In addition, it is often difficult to write down exact solutions for the equilibria or to perform general analytical explorations, except in certain special cases (Kot and Schaffer 1986; Zhou and Kot 2011; Bramburger and Lutscher 2019; Lutscher 2019). This analytic difficulty arises because finding equilibria amounts to finding functions that are fixed points of nonlinear integral equations. Finally, most analyses of IDEs lack information about unstable equilibria, as unstable equilibria do not show up in standard numerical experiments. While an argument may be made that it is more relevant to focus on (asymptotically) stable equilibria, in the interest of a complete characterization of the model it would be useful to observe all equilibria.

In order to circumvent some of these difficulties in analyzing IDE models, a number of tools have been developed to approximate or simplify the original model. Special cases, such as an IDE model with a separable kernel or with the Laplace kernel, may allow for analytical results (Kot and Schaffer 1986; Zhou and Kot 2011; Bramburger and Lutscher 2019; Lutscher 2019). Other tools for analyzing IDE models involve

various approximations of the dispersal kernel, which can be used to approximate steady states and eigenvalues of IDEs, and may be employed for models with one or more species. These methods include average dispersal-success (Van Kirk and Lewis 1997, 1999; Fagan and Lutscher 2006), modified average dispersal-success (Reimer et al. 2016), and geometric symmetrization (Kot and Phillips 2015; Phillips and Kot 2015; Rinnan 2018a, b; Marcinko 2020).

In this paper, we focus on an alternative for analyzing the population dynamics of a single species confined to a finite spatial domain. In contrast to the methods discussed above, which approximate the dispersal kernel, we approximate the growth function. Drawing inspiration from early work by Mark Lewis on the infinite-domain problem (Kot et al. 1996) and more recent developments by Otto (2017) and Nestor and Li (2022), we use a piecewise-constant function to approximate the growth function. We refer to the IDE with this approximated growth as a block-pulse IDE. An m-step block-pulse IDE has m terms, or steps, in the block-pulse approximation. This terminology does not preclude using block-pulse approximations for the dispersal kernel as well; indeed, many applications of block-pulse methods for solving other integral problems involve approximating the entire integrand rather than a single component (Jiang and Schaufelberger 1992; Babolian et al. 2008).

The piecewise-constant nature of the growth function makes the block-pulse IDE simple and analytically tractable. This formulation of the IDE removes many of the analytical barriers mentioned earlier and also offers numerical advantages. With the increased tractability of the block-pulse IDE, we are able to obtain analytic expressions for both the iterates and the equilibria of the model. Of particular interest, we gain explicit formulas for not only the stable but also the unstable equilibria of the block-pulse model.

The analytic formulas for the block-pulse equilibria involve a set of constants whose values are found by using a numerical root-finder to solve a finite, nonlinear set of implicit equations. In other words, we use block-pulse IDEs as part of a hybrid analytical–numerical method for analyzing IDEs. The numerical component, however, is small, and involves solving for a relatively small number of values. Our method is thus more numerically efficient than a pure root-finding method for an operator equation, while the dominant analytic part of the method provides significant insight regarding model equilibria.

Furthermore, this method can be used for a broad set of growth functions, including growth functions with Allee effects. An Allee effect, or depensation, occurs when a population's per-capita recruitment increases, with density, at low densities (Allee et al. 1949; Lewis and Kareiva 1993). Integrodifference equations with Allee effects have, historically, been challenging to analyze (Lutscher 2019). However, the block-pulse method allows us to analyze both simple forms of growth, like compensatory growth, as well as challenging forms of growth, such as depensatory growth.

In Sect. 2, we look at block-pulse series and show how they can be used to approximate growth functions. In Sect. 3, we describe the block-pulse IDE model and discuss assumptions and properties of the model, including a general, closed-form, analytic expression for the iterates of the spatial population-distributions. In Sect. 4, we perform formal analyses of one-, two-, and three-step block-pulse IDEs and determine general patterns for these models. Our analyses include equilibrium solutions, stability and